

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1413

Parkin 138 kV
February 22, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1413. The Interconnection Customer proposes a 256.9 MW battery energy storage facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on February 29, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 532 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,484,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Tangy – Ceiba Baja 7012 138 KV is overloaded to 190.3% of its 229 MVA emergency rating for the loss of the Bear – Clover 345 KV. The project contributes 6.4% DFAX (16.5 MW) to this overload.

The Carmel 146Th St – Oak Crk 138 KV is overloaded to 172.8% of its 243 MVA emergency rating for the loss of the Bear – Clover 345 KV. The project contributes 6.4% DFAX (16.5 MW) to this overload.

The Carmel 146h St – Ceiba Baja 7012 138 KV is overloaded to 171.4% of its 248 MVA emergency rating for the loss of the Clover – Bear 345 KV. The project contributes 6.4% DFAX (16.5 MW) to this overload.

For the loss of Clover – Bear 345 KV line, the Reach – Bluff City 138 KV loads to 134.2% of its 397 MVA rating. Project DFAX: 5.4%.

Red Hill 345/138 kV Transformer Bank: 117.6% loading (506 MVA) following the outage of the Bear – Clover Substation 345 KV; DFAX = 6.1%.

For the loss of South Crk – Aibonito Pds 138 KV, the Aibonito Pds – Intel 138 KV loads to 117.4% of its 277 MVA rating. Project DFAX: 8.2%.

For the loss of Huben – Cape Horn 138 KV L, the Grs Lyon – Sylvan Lake 138 KV loads to 117.1% of its 245 MVA rating. Project DFAX: 5.3%.

The Michigan Power Lp Substation – Osceola 138 KV L is overloaded to 115.3% of its 431 MVA emergency rating for the loss of the Aspen – Lowmoor 345 KV. The project contributes 5.4% DFAX (13.8 MW) to this overload.

The Lone Tree – Osceola 138 KV is overloaded to 115.0% of its 437 MVA emergency rating for the loss of the Lowmoor – Aspen 345 KV. The project contributes 5.4% DFAX (13.8 MW) to this overload.

For the loss of Bear – Clover 345 KV, the Bear – Oak Crk 138 KV L loads to 111.8% of its 285 MVA rating. Project DFAX: 5.3%.

Sylvan Lake – Huben 138 KV: 109.8% loading (268 MVA) following the outage of the Cape Horn – Huben 138 KV; DFAX = 5.3%.

For the loss of Huben – Cape Horn 138 KV line, the Grs Lyon – Velva Wind Interconnection 138 KV L loads to 107.9% of its 262 MVA rating. Project DFAX: 5.3%.

The Unity – Flowood Substation 138 KV is overloaded to 107.1% of its 345 MVA emergency rating for the loss of the Amargo – Boonsboro 138 KV Line. The project contributes 53.9% DFAX (138.5 MW) to this overload.

The Alligator 500/345kV Autotransformer is overloaded to 102.8% of its 896 MVA emergency rating for the loss of the Thiele Kaolin Reedy Crk Substation – Enka 345 KV. The project contributes 5.7% DFAX (14.6 MW) to this overload.

For the loss of Belagio – Otter Riv 138 KV line, the South Colton – Flowood 138 KV loads to 102.5% of its 242 MVA rating. Project DFAX: 26.4%.

Carson Ice Cogen – Alderwood 138 KV Line: 102.3% loading (318 MVA) following the outage of the S Crk Substation – Wythe 138 KV L; DFAX = 5.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tangy – Ceiba Baja 7012 138 KV	Rebuild line with higher-capacity conductor	\$10,164,000	41% (\$4,159,000)
Carmel 146Th St – Oak Crk 138 KV	Rebuild line with higher-capacity conductor	\$7,342,000	41% (\$3,005,000)
Ceiba Baja 7012 – Carmel 146Th St 138 KV	Rebuild line with higher-capacity conductor	\$7,912,000	41% (\$3,238,000)
Reach – Bluff City 138 KV line	Rebuild line with higher-capacity conductor	\$21,662,000	41% (\$8,937,000)
Red Hill 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$17,887,000	21% (\$3,831,000)
Aibonito Pds – Intel 138 KV	Rebuild line with higher-capacity conductor	\$19,131,000	36% (\$6,788,000)
Grs Lyon – Sylvan Lake 138 KV	Rebuild line with higher-capacity conductor	\$19,409,000	6% (\$1,081,000)
Michigan Power Lp Substation – Osceola 138 KV L	Rebuild line with higher-capacity conductor	\$11,804,000	3% (\$401,000)
Osceola – Lone Tree 138 KV	Rebuild line with higher-capacity conductor	\$6,149,000	3% (\$209,000)
Bear – Oak Creek 138 KV line	Rebuild line with higher-capacity conductor	\$2,586,000	14% (\$359,000)
Sylvan Lake – Huben 138 KV	Reconductor line	\$2,876,000	6% (\$160,000)
Grs Lyon – Velva Wind Interconnection 138 KV L	Reconductor line	\$2,866,000	6% (\$160,000)
Unity – Flowood Substation 138 KV	Reconductor line	\$2,340,000	51% (\$1,195,000)
Alligator 500/345kV Autotransformer	Replace transformer with higher-rated unit	\$31,003,000	2% (\$608,000)
Flowood – South Colton 138 KV	Reconductor line	\$991,000	39% (\$387,000)
Carson Ice Cogen – Alderwood 138 KV	Reconductor line	\$1,572,000	61% (\$966,000)
	Total allocated to project		\$35,484,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.